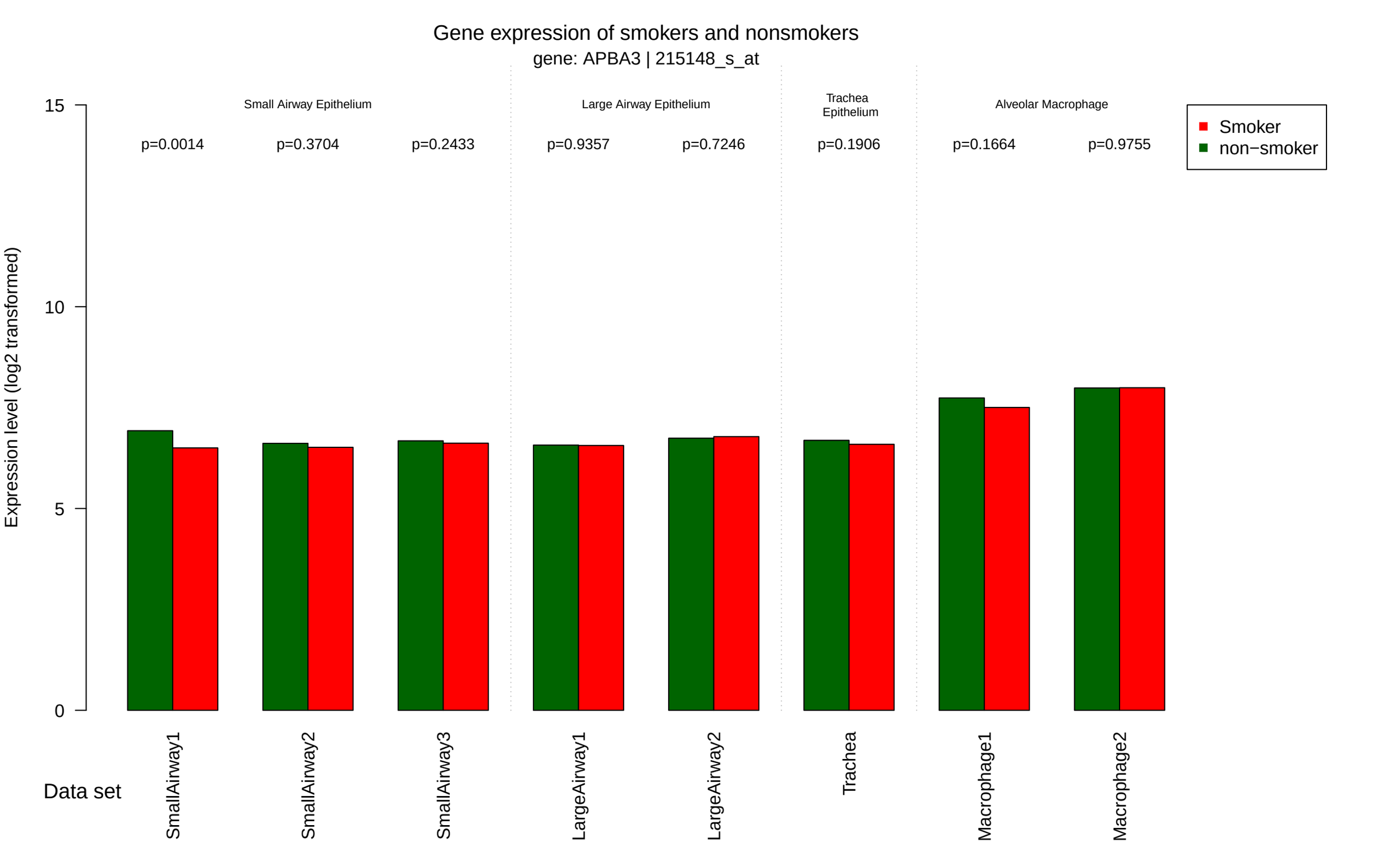
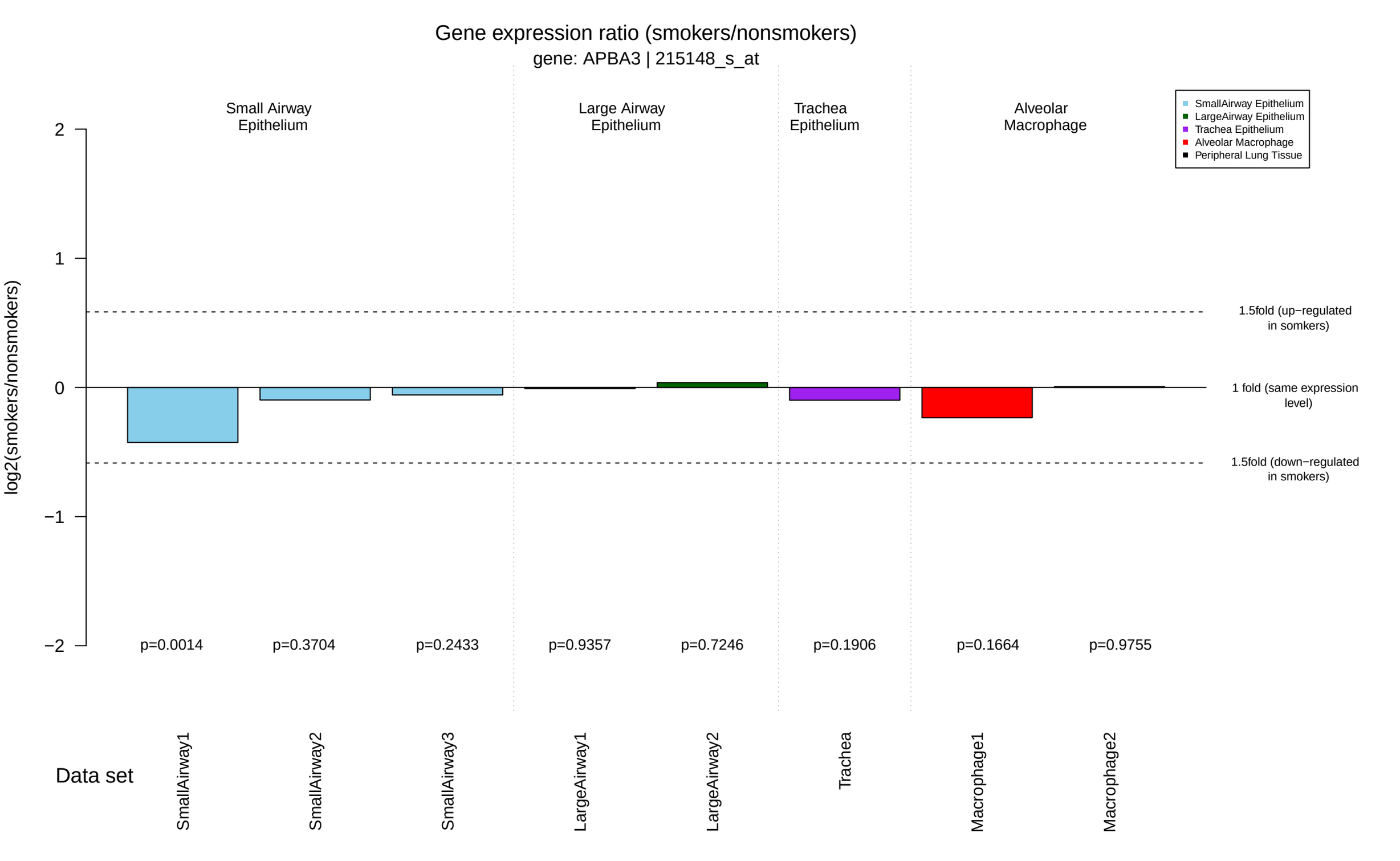
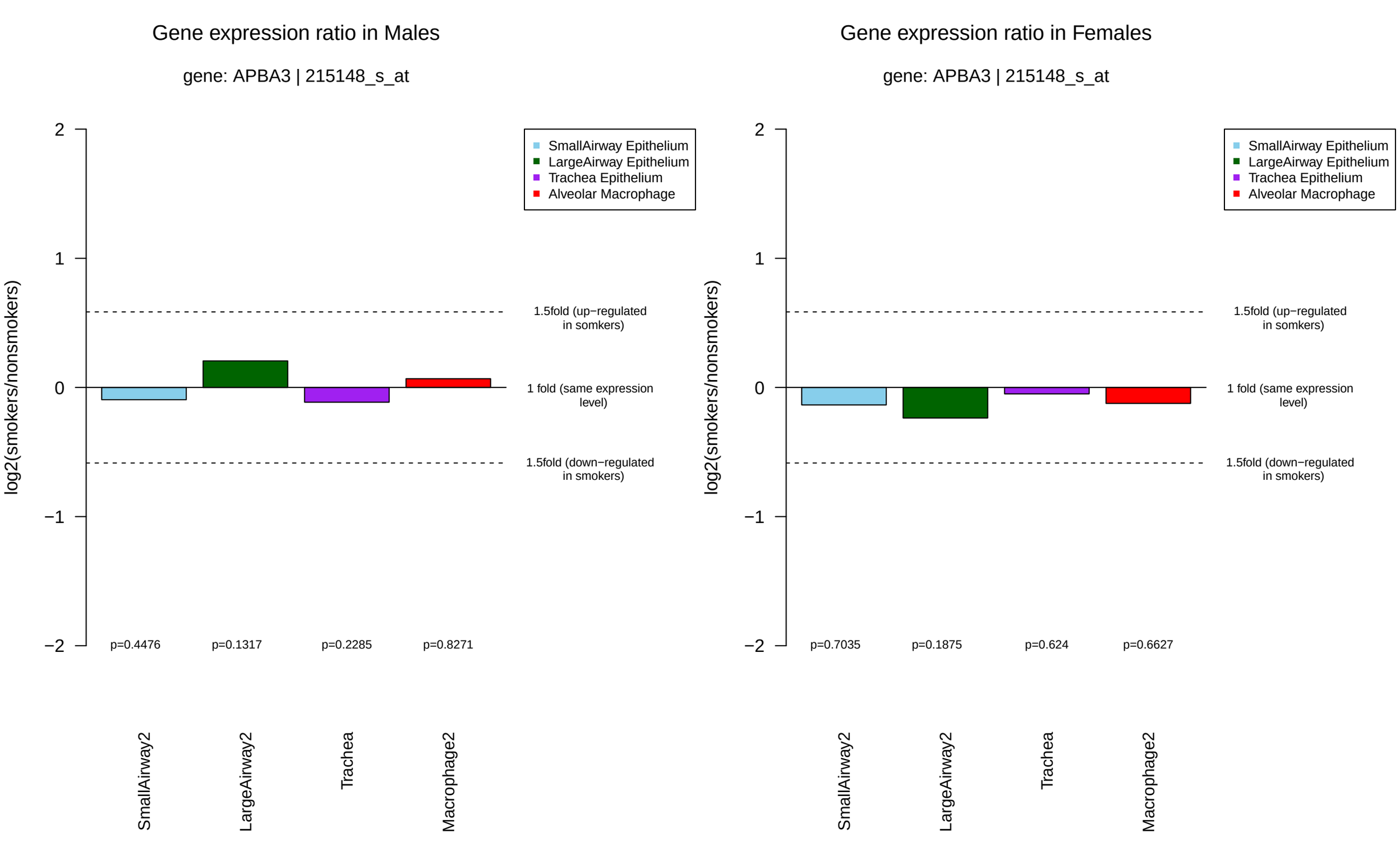
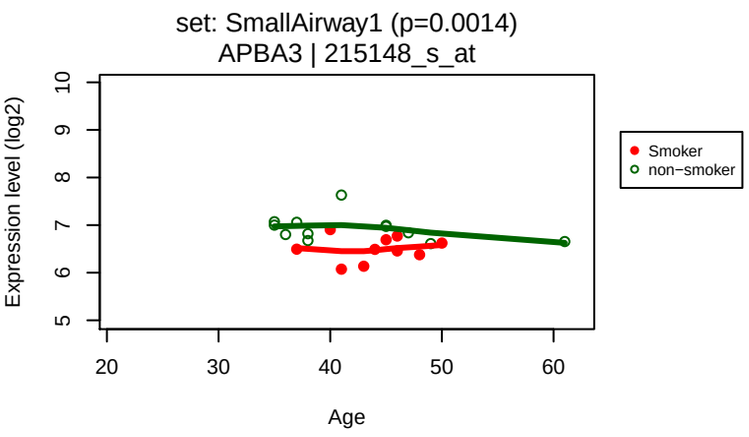
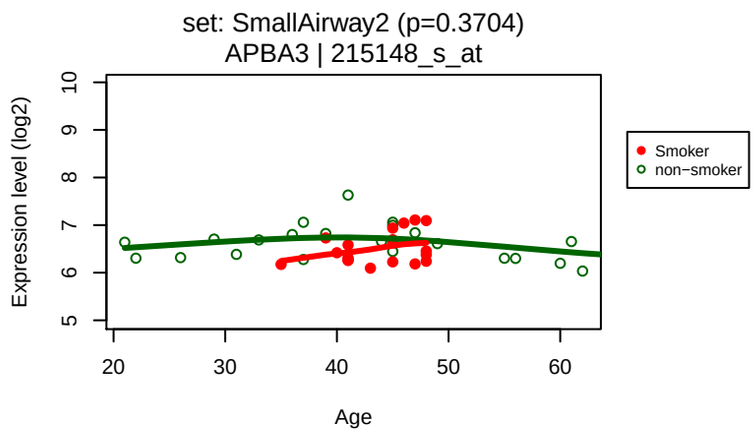
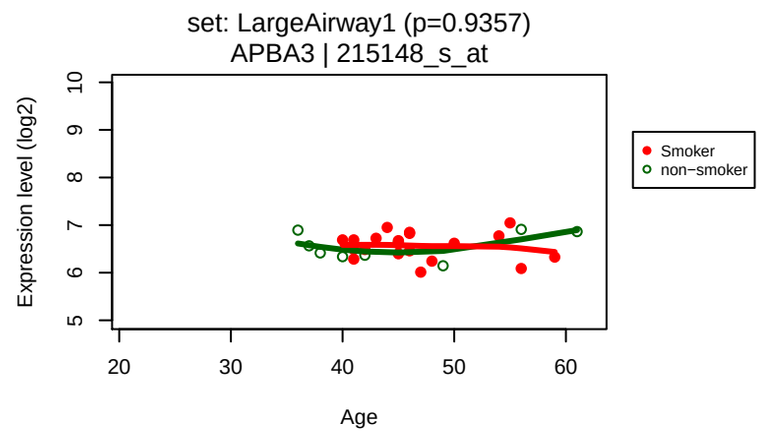

Gene expression of smokers and nonsmokers
gene: APBA3 | 215148_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0014
p=0.3704
p=0.2433
p=0.9357
p=0.7246
p=0.1906
p=0.1664
p=0.9755
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

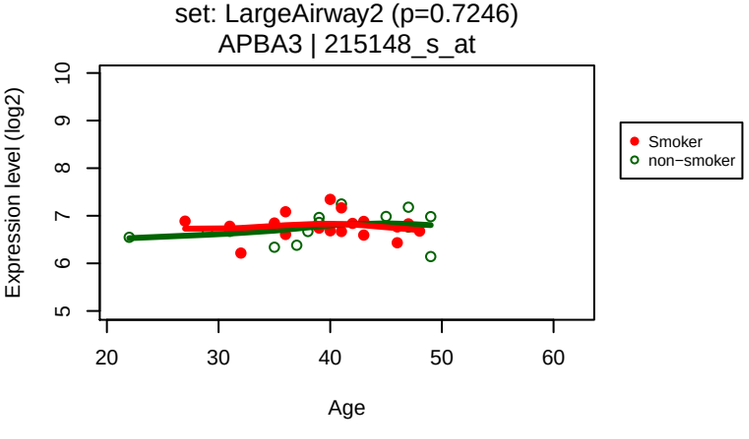

Gene expression ratio (smokers/nonsmokers)
gene: APBA3 | 215148_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0014
p=0.3704
p=0.2433
p=0.9357
p=0.7246
p=0.1906
p=0.1664
p=0.9755
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

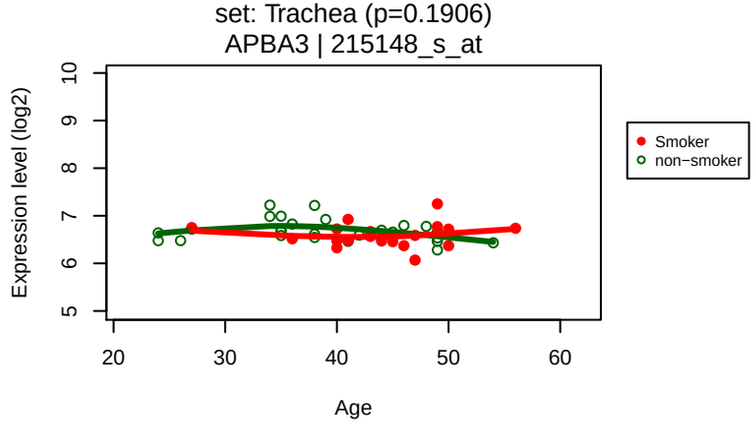

Gene expression ratio in Males
gene: APBA3 | 215148_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4476
p=0.1317
p=0.2285
p=0.8271
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: APBA3 | 215148_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7035
p=0.1875
p=0.624
p=0.6627
SmallAirway2
LargeAirway2
Trachea
Macrophage2

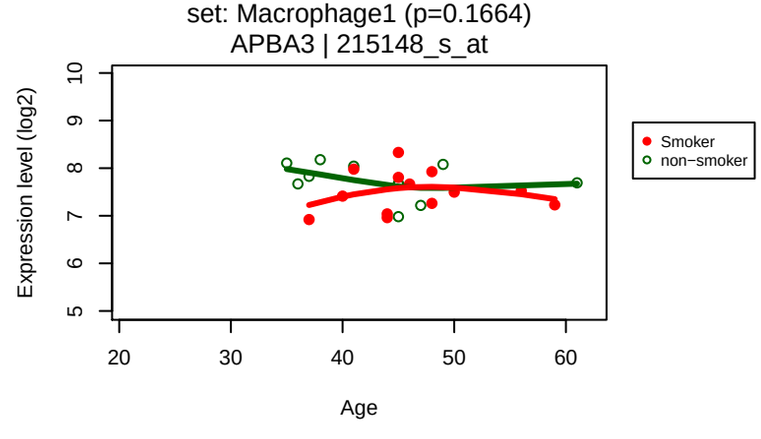

set: SmallAirway1 (p=0.0014)
APBA3 | 215148_s_at
Expression level (log2)
Age
Smoker
non−smoker

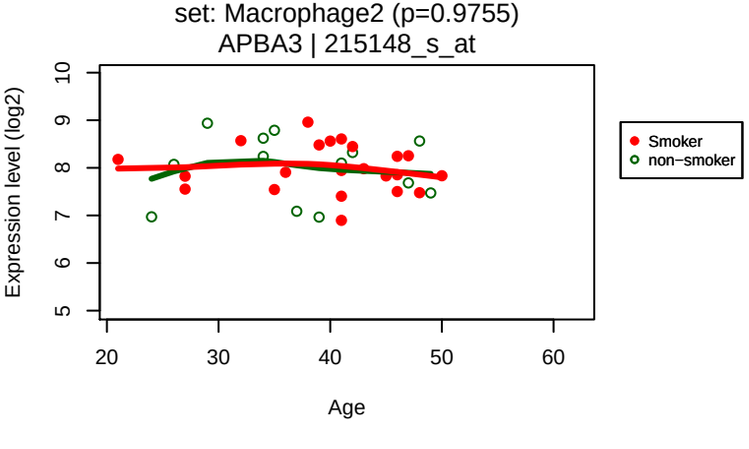

set: SmallAirway2 (p=0.3704)
APBA3 | 215148_s_at
Expression level (log2)
Age
Smoker
non−smoker

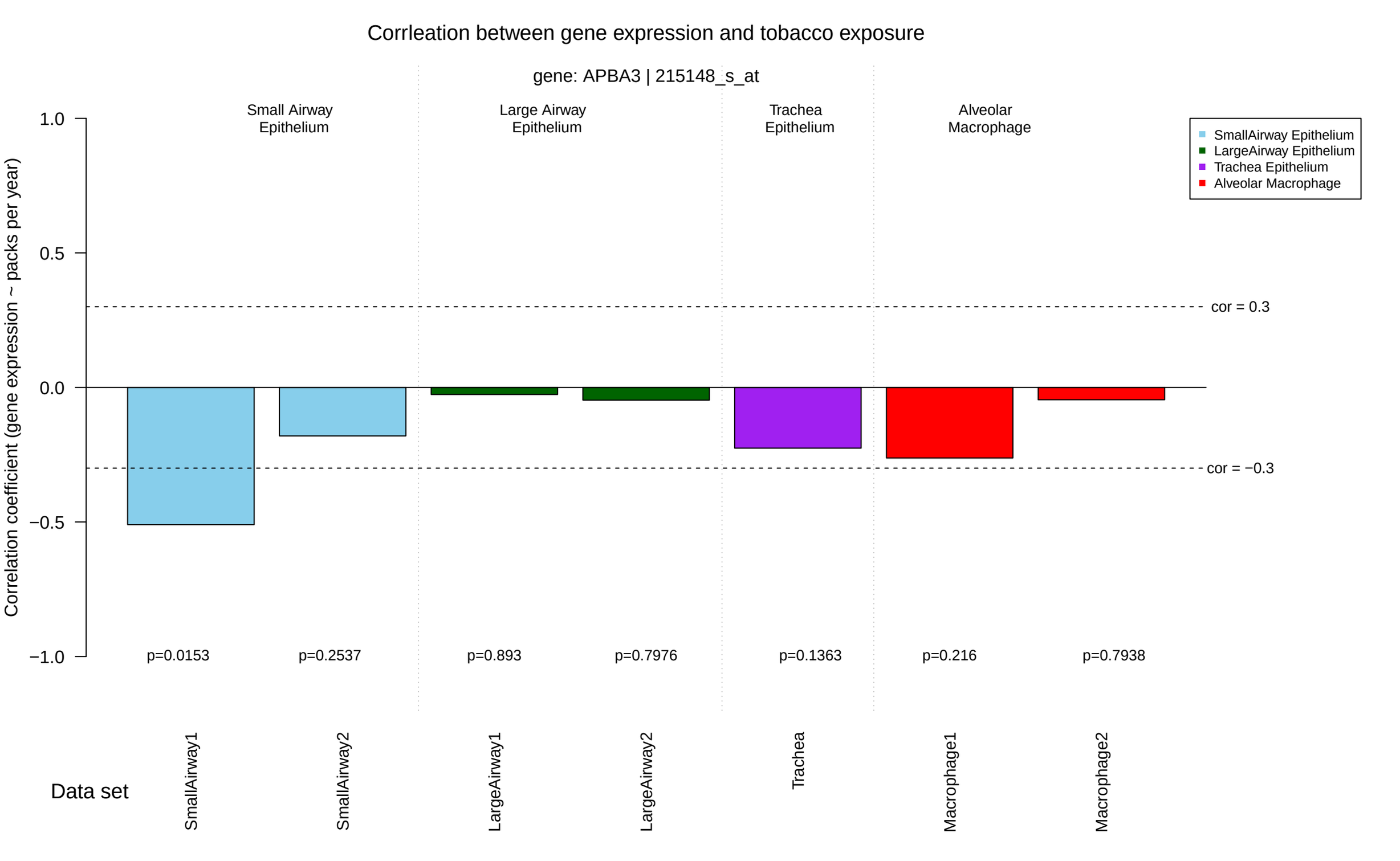

set: LargeAirway1 (p=0.9357)
APBA3 | 215148_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7246)
APBA3 | 215148_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1906)
APBA3 | 215148_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1664)
APBA3 | 215148_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9755)
APBA3 | 215148_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: APBA3 | 215148_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0153
p=0.2537
p=0.893
p=0.7976
p=0.1363
p=0.216
p=0.7938
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2